

# ARE HUMAN SUPER-PREDATORS ALWAYS SUPER-SCARY? A META-ANALYSIS OF WILD ANIMAL BEHAVIOURAL RESPONSES TO HUMAN INTERACTIONS.

Authors: Shawn Dsouza, Kartik Shanker, Maria Thaker  
Centre for Ecological Sciences, Indian Institute of Science, Bengaluru, India.



## INTRODUCTION

Humans interact with wild animals in diverse ways: **lethal** (hunting) and **non-lethal** (tourism, roads).

Our unique ecology makes us efficient, large-scale predators, influencing animal behavior globally.

The relative magnitude of impacts across different type of human interactions have not been quantified.

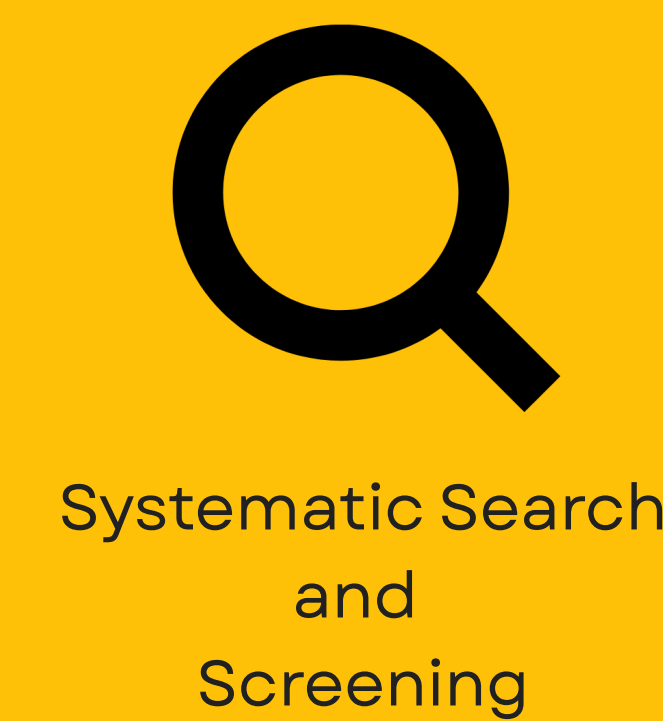
## Question:

How do different types of human interaction influence wild animal **foraging**, **vigilance**, and **movement**?

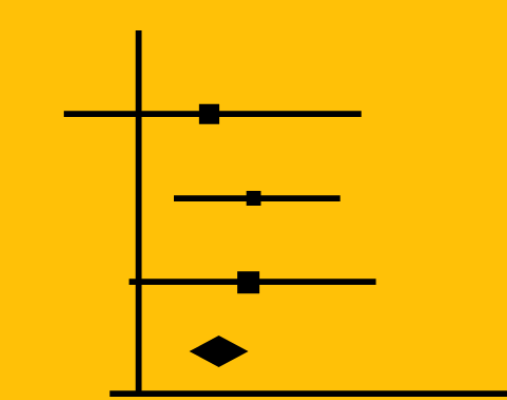
## Hypothesis:



## METHODS



Systematic Search and Screening



Meta-analysis

Search Term: “(human OR anthropo\*) AND (risk OR fear OR NCE OR trait mediated effect OR nonconsumptive\* effect) AND (behavio\*) AND (predat\*)”

Effect size:

Treatment - Control

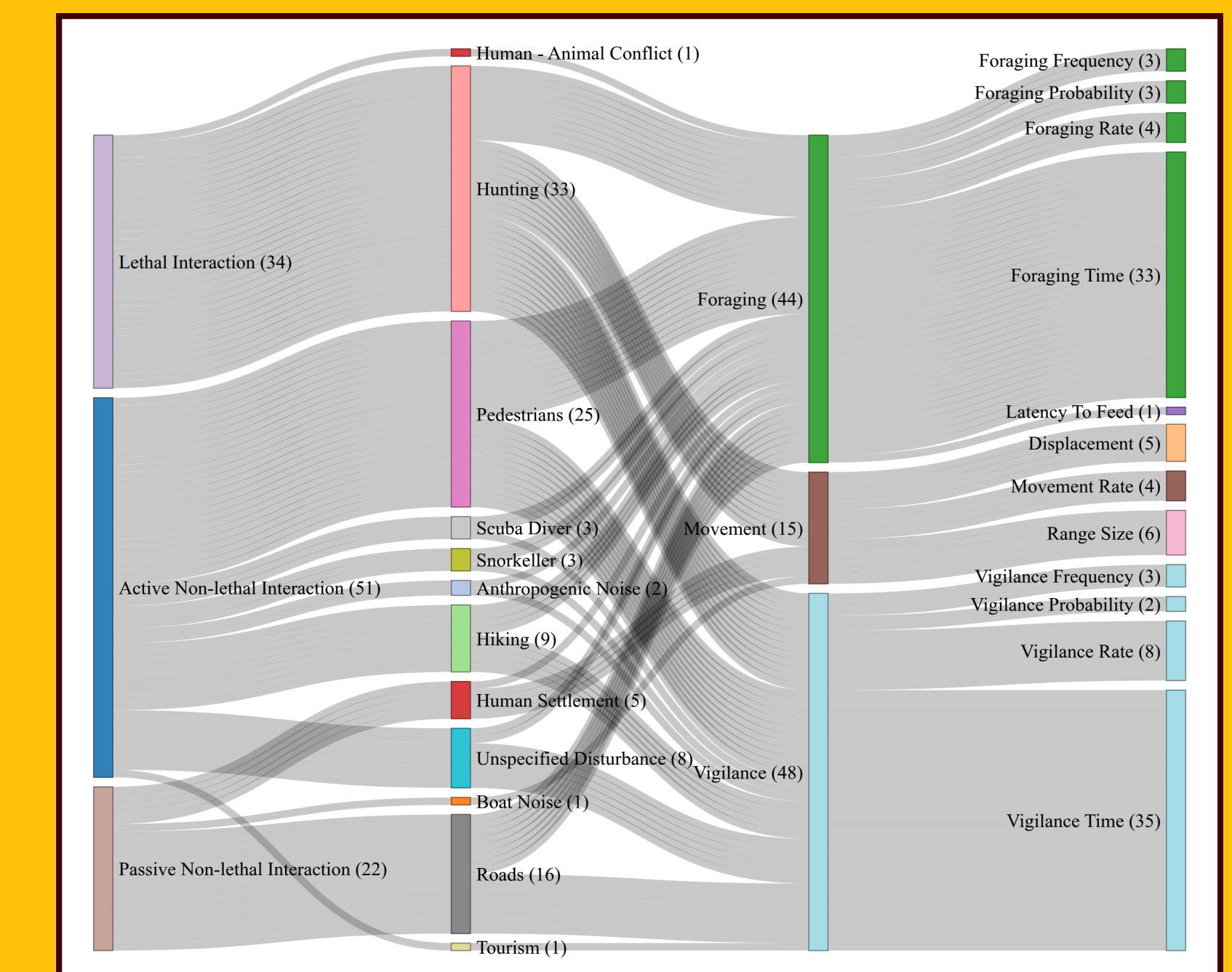
$$d = \frac{Mean_{treatment} - Mean_{control}}{\sigma_{pooled}}$$

Correlational

$$d = \beta_{treatment} \times \sigma_{pooled}$$

Multi-level Meta-analytic model:

$$Observed\ Effect = Type\ of\ interaction + Body\ Size + Random\ Effects$$



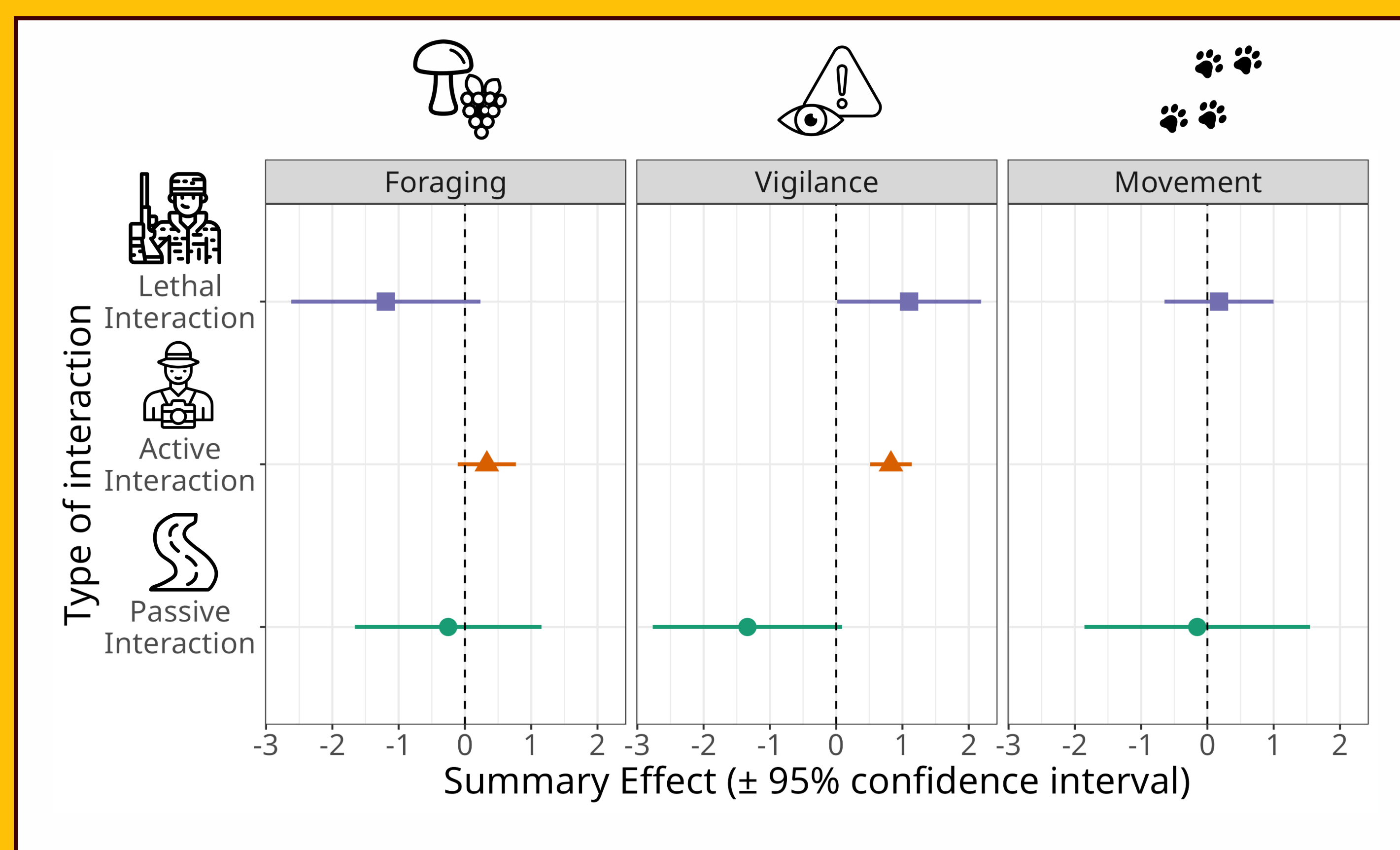
## RESULTS AND DISCUSSIONS

We synthesised ~**30 years** of research including 85 studies (44 in meta-analysis). Our database has a broad taxonomic (**38 species**) and geographic coverage.

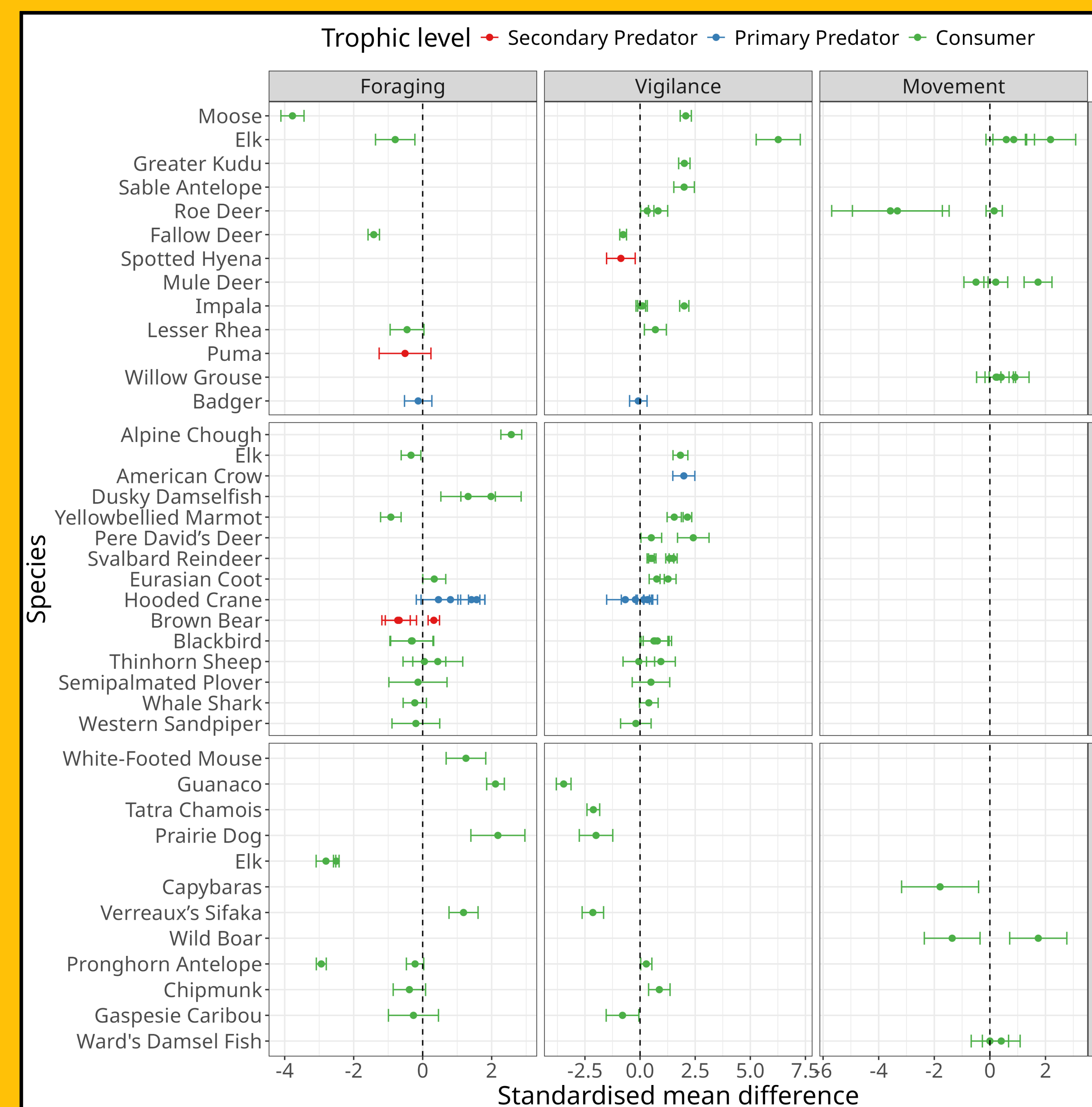
Keyfindings:

**Lethal interactions have the greatest overall effect on wild animal behaviour, followed by non-lethal interactions.**

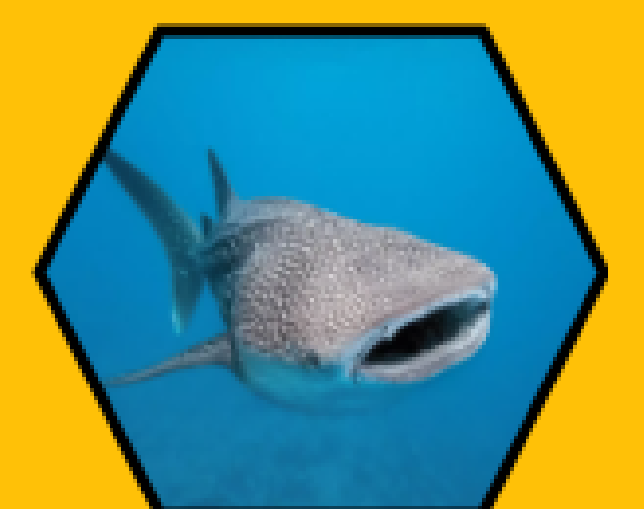
**Human modified spaces may be perceived as safe**



Animal responses to humans in their environment are highly variable indicating **context dependence**.



**Size:** Large animals have a subdued response to humans in their environment.



**History of interaction:** Animals that have a history of lethal interactions such as hunting show large responses to humans.



**Trophic level:** Higher trophic level animals have a subdued response to humans.



**Habituation:** Animals that frequently interact with humans such as urban birds and rodents have smaller responses to interactions.

